Global Reporting Regulation, Stock Liquidity, and Policy Diffusion

Abstract

We examine stock liquidity around 17 families of reporting regulation and their international diffusion. The analysis separates national implementation, issuer coverage and actual public reporting. The market panel contains 1,055,142 security-years in 90 jurisdictions. Source-supported comparisons cover eight first-year families and six balanced-window families. Official statutory filings extend extractive coverage, while original annual reports identify first reporting under the 2010 UK governance-code revision. First-year log-spread contrasts are -0.088 under equal family weights and +0.184 under market-capitalisation weights. Audit reporting receives 84.7% of the latter weight. The exact decomposition shows how policy composition changes the aggregate. Earlier voluntary practices and complementary liquidity outcomes limit a uniform information-benefit interpretation. No diffusion coefficient survives broad multiplicity adjustment. The evidence identifies the importance of policy mix, issuer scope and information timing in comparing reporting mandates.

1. Introduction

Reporting regulation can improve liquidity by supplying credible information and reducing adverse selection. The information increment depends on the covered firms, the audience receiving the report and the disclosure that was already available. We examine liquidity around reporting mandates and ask whether prior foreign implementation helps explain policy diffusion. Aggregation poses a further question: whether an estimate represents an average regulation family or the regulatory exposure of large issuers.

Following the empirical logic of Krueger et al. (2023), we establish the reporting obligation and public-information date before connecting them to issuer outcomes. The setting extends beyond comprehensive sustainability reporting to 17 economic regulation families. We then apply the competition, learning, emulation and coercion framework in Koning, Mertens and Roosenboom (2018). Geographic exposure and source-supported issuer comparisons are reported separately.

Across eight supported first-year families, log-spread contrasts are -0.088 with equal family weights and +0.184 with market-capitalisation weights. The latter assigns 84.7% to audit reporting. We decompose the difference exactly and report every single-family exclusion. The source evidence also distinguishes first required reporting under a revised governance code from first-ever governance disclosure. These comparisons establish the population and policy mix to which each estimate applies; they do not establish a common causal reporting benefit.

2. Institutional setting and research design

A public report and confidential CbCR submitted to tax authorities are different information events. Comprehensive mandatory E/S/G requires a response on environmental, social and governance topics, with the last topic determining the joint date. Climate-only rules, CSR spending and voluntary guidance are classified separately. The inherited topic review retains 37 qualifying phases from 41 country reviews; earliest topic histories remain incomplete. Bank public CbCR attaches to institutions and consolidation perimeters, not automatically every listed holding company.

Eligibility and timing are rule specific (Table 1). Companies House identifies Rio Tinto plc's FY 2015 statutory extractive filing on 29 June 2016. Each of its four downloadable CSVs confirms that date. The voluntary Taxes paid release on 28 June remains a separate event. Equivalent official filings identify Anglo American on 30 June and Ferrexpo on 28 June. Fresnillo's full statutory report is public in an issuer announcement on 15 June; its registry filing day remains unknown. These UK parents are connected to historical ordinary shares and original all-data reports; Rio's Australian sibling is not counted as another treated issuer.

The governance study examines the 2010 UK Code revision, applicable to financial years beginning on or after 29 June 2010. BAE, Pearson, GKN, Drax and ITV explicitly report against it for FY 2011, public in 2012. Explained departures at Pearson and GKN remain covered responses. Earlier Combined Code reports and early adoption are retained: Drax had already adopted annual director re-election. BAE's dated publication announcement is 26 March 2012; ITV's full report appears in its 29 March announcement. Other cases retain year precision where approval, mailing or submission dates do not establish the first public day. This is a revised-code reporting cycle, not initial UK governance adoption.

The 2000-2025 panel contains 99,100 vendor quote/company identifiers. Outcomes are log median relative quoted spread, log median Amihud price impact and the fraction of valid observed returns equal to zero. Each requires 120 valid observations; logged proxies use annual 1/99 winsorisation. Calendar-year measures and observed-return denominators differ from the reference study. Exact bounded historical share identifiers corroborate the selected source matches; current names alone do not establish historical legal identity.

Within each family and cohort, we subtract the weighted comparison change from the implementer change relative to g -1. Geographic designs require known later implementation, 30 issuers per country and three comparison markets. Source-supported designs require three treated and 20 comparison issuers. First-year contrasts use g -1 and g; balanced support is fixed from g -3 through g +2 and averages years 0-2. Unknown histories are not coded never-treated.

GDP in g -1 is allocated over each fixed source-study country issuer universe, and cohorts are weighted by treated mass. Across families, equal weights assign 1/F and market-cap weights use pre-event treated capitalisation. Both use the same supported family set for all three outcomes; banking enters once. Equal-issuer and issuer-cap within-family alternatives are disclosed. Repeated issuer-policy exposures are not a unique-issuer investment portfolio.

US comparisons use contemporaneous parent-incorporation evidence. Governance and extractive studies additionally require US listing evidence and withhold direct foreign-regime flags; IFRS requires auditor US GAAP language. Governance candidates are the 120 largest available nonfinancial parents by pre-event capitalisation. All source and outcome failures remain visible; subsidiary exposure, US governance changes and industry differences remain confounds. Country-sampling diagnostics accompany geographic exposure; covered-study intervals describe issuer sampling with shared-issuer covariance, not causal policy uncertainty. Sparse common legal waves and pretrends constrain identification.

3. Liquidity, report timing and policy composition

Table 2 Panel A reports geographic exposure. Fifteen first-year families yield equal-family spread -0.049 and price impact -0.072, compared with +0.015 and +0.009 under market-cap weights. Confidential tax CbCR receives 33.1% of that exposure weight. Nine families support the balanced window. Pre-policy differences and incomplete eligibility prevent interpreting these broad contrasts as effects on every legally covered issuer.

Panel B uses eight source-supported first-year families. Equal-family spread is -0.088; market-cap spread is +0.184. Corresponding price-impact estimates are +0.013 and +0.188. Figure 1 shows all families and their weights. The market-cap effective family count, 1/sum(w squared), is 1.39. The cap-minus-equal spread difference is +0.272 log points. The decomposition multiplies each fixed family estimate by its weight change. Every single-family exclusion is disclosed; none determines the main specification.

Table 3 identifies the studies. First-year governance support comprises five treated issuers and 40 comparisons. Its spread contrast is -0.155, price impact -0.035 and zero-return fraction +0.00026. Balanced support retains 37 comparisons. Mandatory response to a revision does not establish new information or a new governance practice. The euro-area crisis, UK-US industry differences and contemporaneous US reforms also affect this comparison.

The extractive study includes seven treated issuers and 35 first-year comparisons. The spread contrast is -0.008; price impact is +0.128. Balanced support retains 24 comparisons and spread -0.089. The positive statutory filings resolve parent coverage and dates, but not foreign-subsidiary exposure or the incremental information in earlier voluntary reports. Commodity prices, Brexit and company-specific changes remain competing explanations.

Six families support balanced years 0-2: banks, conflict minerals, IFRS, German nonfinancial, extractive and revised-code governance reports. Spreads are -0.085 and -0.084 under equal and market-cap family weights, while price-impact contrasts are +0.073 and +0.160. Figure 2 reports fixed-sample profiles and pre-policy differences. Changing support prevents joining the eight-family first-year contrast to this six-family path as a single dynamic effect.

4. International diffusion and conclusion

Following Koning et al., competition uses prior implementation among export-pattern peers; learning uses prior outcomes among geographic neighbours; emulation uses language, legal-origin and IGO peers; external dependence uses IMF credit and aid. Signals precede the focal decision. Family-year strata and country-clustered covariance accompany timing and status alternatives, with lagged income, growth and trade controls where specified. The ledger records documented phases rather than certified first-ever adoption.

The full matrix is refitted: 1,330 designs yield 433 stable focal coefficients, none surviving the broad Holm correction. Issuer statutory filings and a governance-code revision do not create new independent national first-adoption decisions. Later liquidity estimates are not inserted into earlier learning signals. Reporting mandates differ in covered economic mass and information increments. Policy composition therefore provides an organising explanation for their aggregate contrasts, while source selection, common legal waves and pretrends limit causal and diffusion conclusions.

Table 1. Implemented regulation families and issuer evidence

Economic regulation family	Geographic support First / balanced	Covered-issuer evidence / remaining limit
Audit oversight / rotation	Yes / Yes	Auditor/client history needed
Expanded auditor reporting	Yes / Yes	US CAM filing cohorts
Beneficial ownership	Yes / Yes	Listed exemptions; venue gaps
Banking public CbCR	Yes / Yes	Bank perimeter; 3 dated publishers
Confidential tax CbCR	Yes / Yes	Nonpublic returns; UPE gaps
General public tax CbCR	Yes / No	No observed eligible filing cohort
Climate disclosure	Yes / Yes	Historical thresholds/report gaps
Sectoral climate reporting	No / No	Sector and panel boundary
Digital assets	Yes / Yes	Crypto object; equity scope differs
Comprehensive mandatory E/S/G	Yes / Yes	3 German statutory report publishers
Extractive payments	Yes / No	7 statutory reporters; group-perimeter gaps
Corporate governance	Yes / No	5 UK revised-Code reporters; earlier practices
IFRS reporting	Yes / No	4 dated Canadian first IFRS reporters
Supply-chain reporting	Yes / Yes	3 US first-cycle reporters
Financial sustainability disclosure	Yes / No	Late phases; actual reports missing
Informed sustainability reporting	No / No	Scope and calendar unsupported
XBRL / digital filing	Yes / No	US early phase; first exhibit unknown

Coverage through 2025 only; future schedules are excluded. Geographic support is outcome support, not verified legal eligibility. Covered issuer studies support eight families in the first year and six in the balanced observed-report subset. Country and issuer thresholds, exemptions, reporting periods and actual publication dates remain separate. Full legal scopes and event-level gap overlays are online.

Table 2. Both cross-family weighting schemes

Population / coverage / outcome	Equal family Estimate [interval]	Market-cap family Estimate [interval]
A. Geographic issuer exposure First year (15 families): Log spread	-0.049 [-0.082, -0.016]	+0.015 [-0.027, +0.056]
A. Geographic issuer exposure First year (15 families): Log price impact	-0.072 [-0.115, -0.030]	+0.009 [-0.059, +0.078]
A. Geographic issuer exposure First year (15 families): Zero-return fraction	-0.0076 [-0.0105, -0.0048]	-0.0025 [-0.0063, +0.0014]
A. Geographic issuer exposure Years 0-2 (9 families): Log spread	-0.032 [-0.088, +0.023]	+0.040 [-0.039, +0.120]
A. Geographic issuer exposure Years 0-2 (9 families): Log price impact	-0.013 [-0.080, +0.054]	+0.095 [-0.025, +0.215]
A. Geographic issuer exposure Years 0-2 (9 families): Zero-return fraction	-0.0018 [-0.0077, +0.0040]	+0.0067 [-0.0073, +0.0206]
B. Source-corroborated subset First year (8 families): Log spread	-0.088 [-0.134, -0.042]	+0.184 [+0.148, +0.221]
B. Source-corroborated subset First year (8 families): Log price impact	+0.013 [-0.075, +0.101]	+0.188 [+0.126, +0.250]
B. Source-corroborated subset First year (8 families): Zero-return fraction	+0.0014 [-0.0029, +0.0057]	+0.0057 [+0.0036, +0.0079]
B. Source-corroborated subset Years 0-2 (6 families): Log spread	-0.085 [-0.168, -0.003]	-0.084 [-0.165, -0.003]
B. Source-corroborated subset Years 0-2 (6 families): Log price impact	+0.073 [-0.037, +0.183]	+0.160 [+0.054, +0.267]
B. Source-corroborated subset Years 0-2 (6 families): Zero-return fraction	-0.0003 [-0.0051, +0.0045]	+0.0009 [-0.0035, +0.0052]

GDP weights within families. Panel A intervals condition on country sampling and the observed policy calendar; Panel B intervals describe issuer sampling only and are not causal policy confidence intervals. Zero returns are fractions. Samples are fixed within each horizon; family composition differs across horizons and populations. Equal-issuer/cap within-family alternatives and full diagnostics are online. No unsupported family is assigned a zero estimate.

Table 3. Covered-issuer first-year comparisons beyond banking

Source-corroborated study	Treated / comparison issuer-cohort observations	Log spread	Log price impact	Zero-return fraction
US auditor CAM (2019-21)	1,481 / 2,767	+0.236	+0.207	+0.0063
US early XBRL (2009)	6 / 62	-0.329	-0.417	-0.0004
US conflict minerals (2014)	3 / 70	-0.085	+0.249	+0.0071
Observed bank full reports (2015)	3 / 21	+0.089	+0.077	+0.0155
Canadian first IFRS reports (2012)	4 / 42	-0.235	+0.132	-0.0096
German statutory E/S/G (2018)	3 / 35	-0.217	-0.235	-0.0124
UK/FR statutory extractive reports (2016)	7 / 35	-0.008	+0.128	+0.0044
UK 2010 Code revision reports (2012)	5 / 40	-0.155	-0.035	+0.0003
Alternative: bank initial phase (2014)	23 / 21	+0.035	-0.343	+0.0006

Counts are issuer-cohort observations; comparison issuers can recur across CAM cohorts. GDP is allocated over the fixed study country universe. The observed-report aggregate uses the first eight rows. The last row replaces banking full reports in a separate sensitivity. Sparse, selected treated groups, industry differences, common waves and subsidiary spillovers limit interpretation; REIT comparisons are industry screened rather than legally certified noncovered. IFRS controls require contemporaneous domestic-parent/US-GAAP evidence; German E/S/G controls can have covered subsidiaries. Full estimates retain balanced and alternative weights.

Source review and timing

Original all-data annual reports establish the five governance responses and prior practices. The source review adds 353 full SEC comparison filings. Official Companies House packages supply twelve CSVs for Rio, Anglo and Ferrexpo, resolving positive parent filing dates. Documents, mirrors and failed downloads are separately recorded. Readable selected sources did not require OCR. Fresnillo’s registry filing day, several actual publication days, historical group perimeters and global thresholds remain unresolved.

References and primary sources

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied working paper; Journal of Accounting Research (2024), doi:10.1111/1475-679X.12548.

Koning, M., G. Mertens and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies. Supplied article.

Companies House Extractives Service. FY2015 statutory packages: Rio Tinto plc 00719885, Anglo American plc 03564138 and Ferrexpo plc 05432915. Official register URLs and preserved payload hashes appear in the companion issuer evidence.

FRC and UK Government. Revised Corporate Governance Code (2010), applicable financial years from 29 June 2010. BAE, Pearson, GKN, Drax and ITV FY2011 annual reports and contemporaneous issuer RNS disclosures; original reports recovered from the all-data EU archive.

Canadian bank first-quarter 2012 releases; BASF, Bayer and BMW FY2017 statutory reports; BP, Total SA and Tullow FY2015 payment disclosures. Earlier voluntary information, report dates and legal obligations are separately recorded.

PCAOB AS 3101; SEC Interactive Data CDI Q105.04; SEC Conflict Minerals Release 34-67716; SEC resource-extraction Releases 34-78167 and 34-90679; inherited bank institution/perimeter review. Full primary sources and gap dispositions accompany the report.

Figure 1. Eight source-supported regulation families

First-year GDP-weighted issuer comparisons. Bars describe conditional issuer sampling, not causal policy confidence intervals. Every supported family appears; banking enters once. Right panel shows pre-event market-cap family weights. Zero-return outcomes are fractions. Small treated groups and different industry/country exposures remain material.

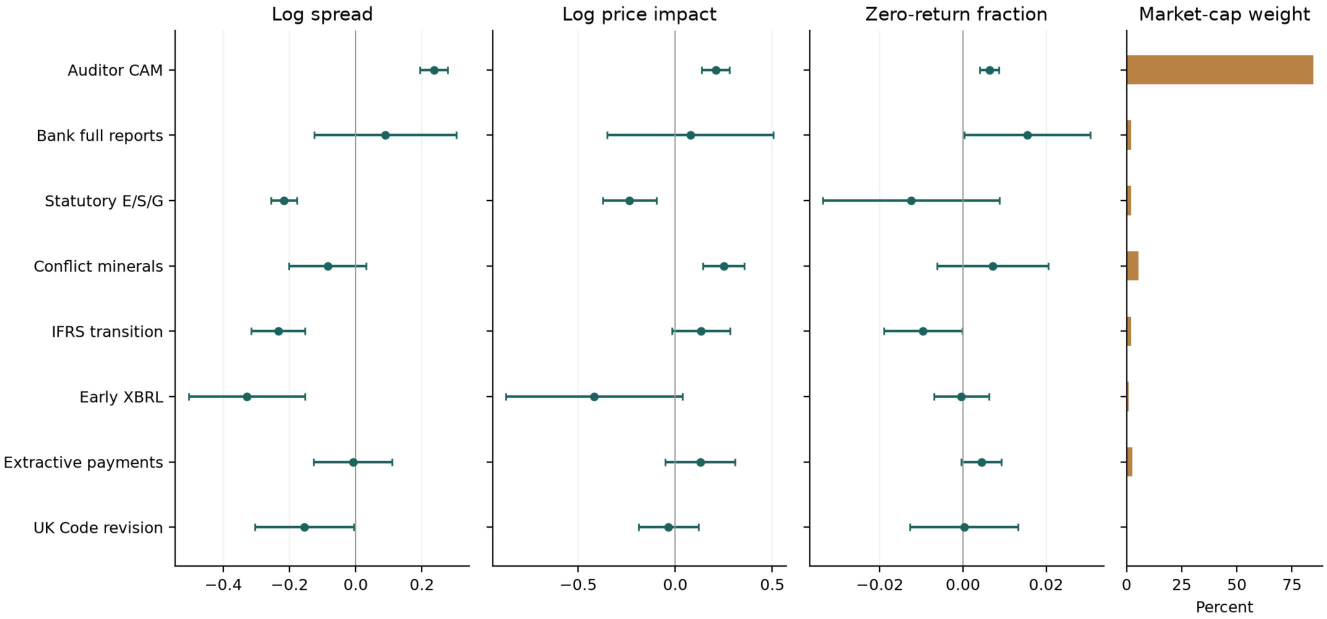
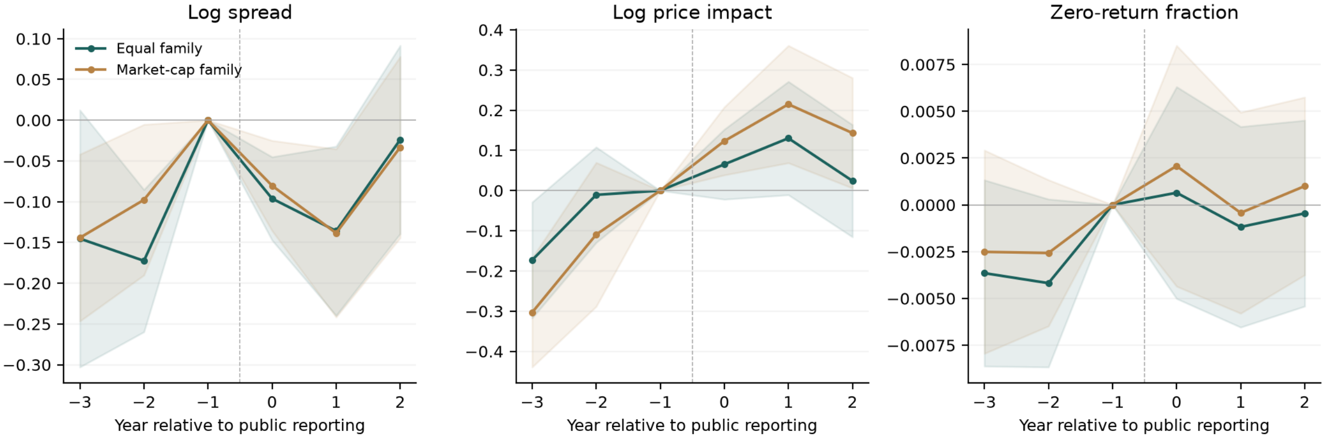


Figure 2. Six-family source-supported balanced profiles

Fixed g-3 through g+2 support: observed bank reports, conflict minerals, Canadian IFRS, German statutory nonfinancial, extractive and UK revised-Code reports. Year -1 is zero. Shading describes pointwise issuer sampling, with shared-issuer covariance retained across families. These sparse collective policy waves do not support causal policy inference. The broad geographic panel has separate adverse lead diagnostics.



Reading the exhibits

The broad exposure panel and the source-corroborated studies answer different questions. The first tracks issuer outcomes around geographic regulation calendars; the second conditions on specific legal or report evidence. The eight-family first-year and six-family balanced estimates differ in both composition and horizon. A favourable geographic coefficient cannot close an eligibility gap. First-year and balanced windows also change support, and should not be joined into one event path.

Reproducible evidence

The companion report provides the frozen design, all family and aggregate estimates, weight contributions, coverage failures, alternative weights, source dispositions and 441 additive event-review JSONs. Licensed issuer and daily market data remain local. The complete diffusion matrix includes failures and coefficients opposing the hypothesised mechanisms. https://hunternbh.github.io/policy-diffusion/